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The Strong No Show Paradoxes are a common flaw in Condorcet voting correspondences

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Abstract. The No Show Paradox (there is a voter who would rather not vote) is known to affect every Condorcet voting function. This paper analyses two strong versions of this paradox in the context of Condorcet voting correspondences. The first says that there is a voter whose favorite candidate loses the election if she votes honestly, but gets elected if she abstains. The second says that there is a voter whose least preferred candidate gets elected if she votes honestly, but loses the election if she abstains. All Condorcet correspondences satisfying some weak domination properties are shown to be affected by these strong forms of the paradox. On the other hand, with the exception of the Simpson-Cramer Minmax and the Young rule, all the Condorcet correspondences that (to the best of our knowledge) are proposed in the literature suffer from these two paradoxes.

1 Introduction

In the theory of voting, the prospects of finding a best voting method have been disappointing, due to the negative results obtained through the systematic axiomatic analysis employed during the last half of this century, including the Arrow Impossibility Theorem and the Gibbard-Satterthwaite result, with its subsequent developments and refinements. We know now that no voting method simultaneously fulfills some minimal properties that apparently are required by any reasonable method, that is to say, no method is free from paradoxes (failures to satisfy some intuitively compelling properties).

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However, it still makes sense to analyze and compare methods in order to select a reasonable one for a given setting. In this task of confronting methods and choosing the right one, perhaps the two main families are the Condorcet and the Positional families.

The interest and relevance of Condorcet voting methods stem from their fidelity to the democratic principle which asserts that if there exists a candidate that is favored by a majority of voters (in a face to face comparison) over any other, this candidate should be the only one chosen. This is called the Condorcet property. For the definition and analysis of the best known Condorcet methods, see Fishburn (1977), Tideman (1987), Laffond et al. (1995), and Peris and Subiza (1999).

The Positional or Scoring methods, and in particular the Borda method, aggregate the preferences of voters through a scoring technique which in some way extracts a measure of the intensity of these preferences. These methods have a normatively appealing consistency property (if two electorates are combined, the global result is coherent with the partial results). For the definition and analysis of the Positional and other related methods, see Young (1974, 1975) and Smith (1973).

Young and Levenglick (1978) have established the incompatibility of the Condorcet and consistency properties. A similar, but independent, property called Participation (none of the voters is disillusioned by submitting his true ballot) has also been shown in Moulin (1988) to be incompatible with the Condorcet property. Hence, every Condorcet method suffers from what has been termed as the No Show Paradox.

This paper, which follows Moulin (1988) and extends some results from Pérez (1995), explores the incidence in Condorcet voting correspondences of two symmetrical strong forms of the paradox (from now on called Strong No Show Paradox, or SNSP for short). In one of these forms, to be called Positive SNSP, there is a voter V_1 whose favorite candidate loses the election if V_1 votes honestly, but wins if V_1 abstains. In the other, Negative SNSP, there is a voter V_1 whose least preferred candidate wins if V_1 votes honestly, but loses if V_1 abstains.

Although not all Condorcet methods suffer from these paradoxes, the Simpson-Cramer Minmax method (free from Positive and Negative SNSP) and the Young method (free from Negative SNSP) are, as far as I know, the only exceptions among those proposed in the literature.

Section 2 presents the basic terminology and some known results. Section 3 defines the Positive and Negative SNSP, and identifies some weak domination properties that imply the paradoxes. Section 4 analyses, with the help of these properties (whenever possible), which known correspondences abide by the paradoxes, and Sect. 5 concludes the paper with some additional remarks, including the analysis of some other versions of SNSP.

2 Terminology

The terminology of Fishburn (1977) and Laffond et al. (1995) will be used whenever possible, with few modifications.

Let $X = \{x_1, x_2, \ldots, x_n\}$ be a finite set with two or more candidates. Preferences of any voter are supposed to take the form of a complete ranking, that is to say, a linear (strict and complete) order L over X. We say $L = xyzt\ldots$ to denote the preference order in which x is the most preferred candidate, y is the second one, and so on, and $a\ L\ b$ means that a is preferred to y in L.

Given the set of candidates $X = \{x_1, x_2, \ldots, x_n\}$, and any finite set $V = \{1, 2, \ldots, m\}$ with one or more voters, we call a **Situation** any pair (X, p), where p is a preference profile over X from V, that is to say, a m-tuple of orders over X, each one meaning the preferences of a voter over X.

We call **Voting Correspondence** (from now on **VC**) any function f which maps any situation (X, p) to a non-empty subset of $X, f(X, p)$. The elements of $f(X, p)$ are the chosen candidates (the winners) over X from the preference profile p. Given any two VCs f and g, we call f a **refinement** of g when $f(X, p) \subseteq g(X, p)$ for every (X, p).

Since we will consider only anonymous VCs (all voters are equally considered), a preference profile over X from V can also be described by specifying how many of the m voters from V sustain any of the $n!$ linear orders on X.

Given any X, any two disjoint sets of voters $V_1 = \{1, 2, \ldots, m_1\}$ and $V_2 = \{m_1 + 1, m_1 + 2, \ldots, m_1 + m_2\}$, and any two preference profiles p_1 and p_2 over X from, respectively, V_1 and V_2, we can merge these two profiles in order to obtain a new profile over X, but now originated from $V_1 \cup V_2$. This new profile will be called $p_1 + p_2$.

Let (X, p) be any situation with n candidates and m voters:

Given any two different candidates x, y from X, $p(x, y)$ means the number of voters in p which prefer x to y. Because ties are not allowed in any voter's ballot, $p(x, y) + p(y, x) = m$. The square nxn matrix M_p, whose entries are $p(x, y)$, will be called the **Comparison Matrix** for (X, p). For every candidate x, the sum of the off-diagonal row entries in M_p is called the **Borda Score** of x.

Candidate x is said to **beat** y, denoted by xW_py, if and only if $p(x, y) > p(y, x)$. If we use $\geq$ instead of $>$, we have the relation x **beats or ties** y, which is denoted by xU_py. Candidate x is said to **beat indirectly** (**beat or tie indirectly**) y, denoted by $xWW_py(xUU_py)$ when there are k candidates $x_1, x_2, \ldots, x_k$ in X such that $x = x_1Rx_2R\ldots Rx_k = y$, being $R = W_p(R = U_p)$. If x beats any other, then x is called the **Condorcet** candidate.

The situation in which the set of candidates is $Z \subseteq X$, and the profile is the restriction to Z of profile p, is called $(Z, p/Z)$. Its comparison matrix is $M_{p/Z}$.

For every order $L : x_1x_2\ldots x_n$ of the candidates, the sum $K(L) = \sum_{i<j} p(x_i, x_j)$ will be called the **Kemeny Score** of L. Given any order $L : x_1x_2\ldots x_n$, we say $\boldsymbol{x}$ **attains** $\boldsymbol{y}$ **through** $\boldsymbol{L}$ in (X, p) if there is a sequence of distinct candidates $a_1, a_2, \ldots, a_j$, with $x = a_1$ and $y = a_j$, such that $a_i\ L\ a_{i+1}$ and $p(a_i, a_{i+1}) \geq p(a_j, a_1)$ for $i = 1, \ldots, j-1$. The order L is called a **stack** of (X, p) if and only if $x_i\ L\ x_j$ implies that x_i attains x_j through L in (X, p).

Let us call **Bipartisan Plurality Game** of (X, p) the two-player symmetric constant-sum game in which X is the set of pure strategies, and the payoffs for the profile of strategies (x, y) are:

If $x \neq y$, $u_1(x, y) = p(x, y)$ and $u_2(x, y) = p(y, x)$

If $x = y$, $u_1(x, y) = u_2(x, y) = m/2$

In order to completely relate the terms $p(x, y)$ and the payoffs of this game, all the entries $p(x, x)$ of the main diagonal will be set as equal to $m/2$. This convention does not significantly affect any previous concept or result, and facilitates some definitions and computations. Thus, M_p is the payoff matrix of the row player.

Any change in M_p by which a unit is added to the off-diagonal entry $p(x, y)$ and subtracted from $p(y, x)$, is called an **elemental interchange in M_p**. Any change in (X, p), by which two consecutive candidates x and y in a voter's order interchange their position in that voter's order, is called an **elemental interchange in (X, p)**.

A VC f is said to satisfy **Translation Invariance** if, for any two situations (X, p) and (X, q), $(q(x, y) = q(y, x)\ \forall x, y \in X)$ implies $(f(X, p) = f(X, p + q))$. That is to say, adding a group of voters whose preferences, as measured by its comparison matrix, are the same for every candidate, does not change the set of winners.

Following Fishburn (1977), we will distinguish three types of VCs. The correspondence f is said a **C1**, **C2** or **C3** Correspondence if, respectively:

C1: For every situation (X, p), the set $f(X, p)$ depends only on the W_p relation.

C2: f is not a C1-Correspondence and for every situation (X, p), $f(X, p)$ depends only on the Comparison Matrix M_p.

C3: f is neither a C1-Correspondence nor a C2-Correspondence.

Definition 1. *A VC f is called **Condorcet** if and only if for every situation (X, p),*

$$(x W_p y\ \forall y \in X \backslash \{x\}) \text{ implies } f(X, p) = \{x\}.$$

That is to say, if there is a Condorcet candidate, it will be the only winner.

Definition 2. *A VC f satisfies the **Consistency** property if and only if for any two situations (X, p_1) and (X, p_2), $f(X, p_1) \cap f(X, p_2) \neq \varnothing$ implies $f(X, p_1 + p_2) = f(X, p_1) \cap f(X, p_2)$.*

In other words, if some candidates are chosen for profile p_1 and profile p_2, they, and only they, are chosen when the two profiles are merged. This property characterizes, along with Anonymity and Neutrality, the positional choice correspondences, whose best known examples are the **Plurality rule** (the winners are those who are the most preferred candidates by a highest number of voters) and the **Borda rule** (the winners are those who obtain the highest Borda Score). See Young (1975), and see also Young (1974) for a characterization of the Borda rule where the Consistency property plays a fundamental role.

The following property, from Moulin (1988), is defined in the context of Voting Functions (VCs which, for any situation, chose only one candidate), which he calls Voting Rules.

Definition 3. *A Voting Function f satisfies the* ***Participation*** *property if and only if for any given pair of situations* (X, p) *and* (X, v)*, where profile v has only one voter,*

$$(f(X, p) = \{x\} \text{ and } x \text{ is preferred to } y \text{ in } v) \text{ implies } f(X, p + v) \neq \{y\}.$$

In words, if x is the winner for a situation and a new voter who prefers x to y is added, candidate y will not become the winner. Thus, the new voter would do better if he abstained, because submitting his ballot would result in the election of a less preferred candidate. Failing to satisfy Participation means that the **No Show paradox** sets in.

The incompatibility of the above two properties with the Condorcet property is shown in Propositions 1 and 2 below, established respectively in Young and Levenglick (1978) and Moulin (1988).

Proposition 1. *No Condorcet VC satisfies the Consistency property.*

Proposition 2. *No Condorcet Voting Function satisfies the Participation property.*

Consistency and Participation are not logically related. Moreover, Moulin (1988) proved that Participation does not imply nor is implied by the Reinforcement property, which is a natural translation of Consistency to the Voting Functions framework.

The following definition is a natural translation of the Participation property to the Voting Correspondences framework.

Definition 4. *A VC f satisfies the* ***VC-Participation*** *property if and only if for any given pair of situations* (X, p) *and* (X, v)*, where profile v has only one voter,*

$$\begin{aligned}&\text{If } x \in f(X, p) \text{ and } x \text{ is preferred to } y \text{ in } v,\\ &\qquad \text{then } (y \in f(X, p + v) \text{ implies } x \in f(X, p + v)).\end{aligned}$$

In other words, if candidate x is chosen for a situation and a new voter is added who strictly prefers x to y, candidate y will not be chosen if she is not accompanied by x.

An easy adaptation of the proof (ii) in Moulin's statement (1988, p. 57–59), as made in Pérez (1995), allows to establish the following proposition:

Proposition 3. *No Condorcet VC satisfies the VC-Participation property.*

3 The Strong No Show Paradoxes

The following properties may be seen as the minimum to require concerning the coherence in the winning set when new voters are added. The first one can be easily shown to be a weakening of both VC-Participation and Consistency.

Definition 5. *A VC f satisfies the* ***Positive Involvement (PI)*** *property if and only if for any given pair of situations* (X, p) *and* (X, v)*, where profile v has only one voter,* *If* $x \in f(X, p)$ *and* x *is preferred to any* y *in* v, *then* $x \in f(X, p + v)$.

We say f satisfies the **Negative Involvement (NI)** *property when*

If $x \notin f(X, p)$ *and any* y *is preferred to* x *in* v, *then* $x \notin f(X, p + v)$.

In other words, Positive (Negative) Involvement requires that if candidate x is chosen (not chosen), x will remain chosen (not chosen) when a new voter is added who prefers x to any other candidate (any other candidate to x).

The failure by a VC f to satisfy any of these properties means that f suffers from what we will call **Strong No Show Paradox** (**SNSP**). We will say that f suffers from the **Positive SNSP** when f fails to satisfy PI, and from **Negative SNSP** when f fails to satisfy NI.

Smith (1973) uses PI to characterize some multistage VCs, Fishburn and Brams (1983) coin the term "No-Show Paradox" and show an example of Negative SNSP for the case of the well-known Preferential Voting, and Saari (1989) shows other examples of the paradox. Concerning Condorcet VCs, Richelson (1978, 1980) shows that the Copeland, Dogdson, Black and Nanson correspondences fail to satisfy PI (there called Voter Adaptability), and Saari (1995) shows that VCs defined by sequential pairwise comparisons according to a specified agenda also fail to satisfy the last property. Pérez (1995) analyses PI (there called Weak Monotonicity in the face of new voters) in the context of Condorcet VCs.

3.1 Impossibility result for some families of Condorcet VCs

The following domination properties will be needed to identify families of Condorcet VCs that fail to satisfy Positive and Negative Involvement.

Definition 6. *Given a situation* (X, p) *and two candidates* x *and* s, *we say* s ***is C1-dominated by*** x *if and only if the two following conditions hold:*

a) $p(x, s) > p(s, x)$
b) *For any* $z \in X \backslash \{x, s\}$, *if* $p(s, z) \geq p(z, s)$ *then* $p(x, z) > p(z, x)$.

In other words, x beats s and x beats any candidate z beaten by, or tied with, s.

Note. If we consider only the information conveyed by the W_p relation (thus focusing on the underlying Tournament structure of the situation), we can say:

1) The C1-domination relation coincides with the covering relation defined in the context of strict tournaments, in which W_p is a complete relation ($\forall x, y$, if $x \neq y$ then $x W_p y$ or $y W_p x$). See Fishburn (1977) and Laffond et al. (1995).

2) The C1-domination relation is stronger than any of the covering relations defined in the context of weak tournaments, in which W_p is the asymmetric part of a complete relation. Thus, if s is C1-dominated by x then s is covered by x. See Peris and Subiza (1999).

Definition 7. *Given a situation* (X, p) *and two candidates* x *and* s, *we say that* s ***is C2-dominated by*** x *if and only if the two following conditions hold:*

a) $p(x, s) > p(s, x)$
b) *For any* $z \in X \backslash \{x, s\}, p(x, z) \geq p(s, z)$.

In other words, x beats s, and x performs equal or better than s in the matrix M_p, in her confrontation with any other candidate. Both C1 and C2 domination concepts are generalizations of the Pareto Domination relation.

Definition 8. *Given a situation* (X, p) *and two candidates x and s, we say that s* ***is C2-quasidominated in differences by*** *x if and only if the three following conditions hold:*

a) $p(x, s) > p(s, x)$.
b) $p(x, z) \geq p(s, z)$ *for any* $z \in X - \{x, s\}$ *except perhaps for a unique* z.
c) *If* $p(x, z) < p(s, z)$, *then* $p(x, s) - p(s, s) > p(s, z) - p(x, z)$.

In other words, x beats s, x performs equal or better than s in her confrontation with any other candidate, except perhaps with only one, say the z candidate, and the difference in favor of x in her confrontation with s (as expressed by the difference $p(x, s) - p(s, s)$) more than compensates for the difference in favor of s when both are confronted with z. Candidate z can be called the *weak point* of x with respect to s.

Definition 9. *Let* (X, p) *be any situation. Given three different candidates x, y and s, we say that s* **is C2-dominated by the pair** $\{x, y\}$ *if and only if the two following conditions hold:*

a) *Both x and y C2-quasidominate s in differences.*
b) *If* $w \in \{x, y\}$ *and* $p(w, z) < p(s, z)$, *then* $p(y, z) - p(s, z) \geq p(s, z) - p(x, z)$.

That is to say, besides the fact that both x and y C2-quasidominate s, the performance of any of them at the weak point of the other is enough to compensate the poor performance of the other at its own weak point. This compensation causes that, in every column of M_p, the entry corresponding to candidate s is equal or lower than the average of the entries corresponding to candidates x and y.

Although stronger, this definition is formulated in the spirit of the concept of weak domination of a pure strategy by a mixed strategy in finite strategic-form games. In fact, in the Bipartisan Plurality Game associated to a situation (X, p), defined in Laffond et al. (1993, 1994), if a candidate s is C2-Dominated by the pair $\{x, y\}$, then the pure strategy s is weakly dominated by the mixed strategy $0.5x + 0.5y$.

Every concept of domination among candidates tell us that, from the perspective of this concept, a dominated candidate, being surpassed by other(s), does not perform sufficiently well in the preferences of voters and, therefore, does not deserve to win the election. So, for every concept of domination, it is relevant to pose the question of which VCs respect that concept, by not electing as a winner the dominated candidate or, at least, by not electing him as the unique winner. The following definition, being of a general character, is applicable to the four concepts of domination as defined above.

Definition 10. *A VC f* ***Weakly Respects the Q-Domination*** *if and only if for any given situation (X, p), s is Q-Dominated implies that $f(X, p) \neq \{s\}$. (We say f* ***Respect the Q-Domination*** *if the consequent of the implication is $s \notin f(X, p)$).*

There is an obvious relation between the just defined properties of a VC and its refinements. Indeed, for any Q-domination concept, if f is a refinement of g, the two following statements hold:

a) *If g respects the Q-Domination so does f.*
b) *If f weakly respects the Q-Domination so does g.*

The following, the main proposition of the paper, establishes a logical incompatibility between PI (or NI with Translation Invariance) and the property of weakly respecting some of the above defined domination concepts. Part (a) is a strengthening of Proposition 5 in Pérez (1995).

Proposition 4.

a) *No Condorcet VC that weakly respects the C1-Domination or weakly respects the C2-Domination by a pair, satisfies the PI property.*

b) *No Condorcet VC that weakly respects the C1-Domination or weakly respects the C2-Domination by a pair, satisfies the NI and the Translation Invariance properties.*

Proof. We will need the following lemma.

Lemma 1. *Given any Condorcet VC f, any situation (X, p) and any two candidates x and z, if f satisfies PI or f satisfies NI and Translation Invariance, then $p(x, z) < \mathrm{Min}_{y \in X} p(z, y)$ implies $x \notin f(X, p)$.*

Proof of the Lemma. The case in which f satisfies PI has been proved in Pérez (1995, p. 146), by an easy adaptation of the proof of a similar result in Moulin (1988, p. 57). To prove the case in which f satisfies NI and Translation Invariance, let m be the number of voters in profile p, let $X = \{x_1, x_2, \ldots, x_n\}$ with $x = x_1$ and $z = x_2$, suppose $p(x, z) < \mathrm{Min}_{y \in X} p(z, y)$ and let $h = p(z, x) - \mathrm{Min}_{y \in X} p(z, y)$. If z is the Condorcet candidate, x can not be chosen in (X, p). Suppose z is not Condorcet. Let us add to the profile p, for every permutation σ of $\{1, 2, \ldots, n\}$, h voters with preferences $x_{\sigma(1)} x_{\sigma(2)} \cdots x_{\sigma(n)}$. It is obvious that, in the new profile p' with $m + h(n!)$ voters, every entry $p'(x, y)$ satisfies $p'(x, y) = p(x, y) + h(n!)/2$. Hence, $p'(x, z) < \mathrm{Min}_{y \in X} p'(z, y)$ and $h = p'(z, x) - \mathrm{Min}_{y \in X} p'(z, y)$. If h voters, all with identical preference order $\ldots zx$, are now removed from p', z will become the Condorcet candidate for the new profile p''. Indeed, in this removal process no non-diagonal entry of the x row, and only the $p'(z, x)$ entry of the z row, decrease. Hence, $\mathrm{Min}_{y \in X} p''(z, y) = p''(z, x) = p'(z, x) - h = \mathrm{Min}_{y \in X} p'(z, y)$, and $p''(x, z) = p'(x, z) < \mathrm{Min}_{y \in X} p'(z, y) = \mathrm{Min}_{y \in X} p''(z, y) = p''(z, x)$. Candidate z is thus the Condorcet candidate and the only chosen for (X, p''). Hence, x can not be chosen for (X, p), because in that case x would also be chosen (by Translation Invariance) for (X, p'), and would also be chosen (by NI) for (X, p''). □

To complete the proof of proposition 4, let $X = \{x, y, z, u, t\}$ and p the following profile: [*yxtuz* (11 voters), *uzytx* (10 voters), *xztyu* (10 voters), *uztyx* (2 voters), *utzyx* (2 voters), *zyxtu* (2 voters), *tzyxu* (1 voter), *xytuz* (1 voter)]. The comparison matrix is:

$$M_p: \quad \begin{array}{c|ccccc} & x & y & z & u & t \\ x & 19.5 & 11 & 22 & 25 & 24 \\ y & 28 & 19.5 & 12 & 25 & 24 \\ z & 17 & 27 & 19.5 & 13 & 24 \\ u & 14 & 14 & 26 & 19.5 & 14 \\ t & 15 & 15 & 15 & 25 & 19.5 \end{array}$$

Suppose f is Condorcet and satisfies PI or satisfies NI and Translation Invariance. From Lemma 1 applied to the pairs (x, y), (y, z), (z, u) and (u, t), candidates x, y, z, and u are not chosen for (X, p), and thus candidate t is the only winner.

However, both x and y C1-Dominate t and the pair $\{x, y\}$ C2-Dominates t (y is the weak point of x and z is the weak point of y. Moreover, the difference in favor of t at the weak point of x is $p(t, y) - p(x, y) = 4$, while $p(x, t) - p(t, t) = 4.5$ and $p(y, y) - p(t, y) = 4.5$; in a similar way, the difference in favor of t at the weak point of y is $p(t, z) - p(y, z) = 3$, while $p(y, t) - p(t, t) = 4.5$ and $p(x, z) - p(t, z) = 7$). Therefore, f fails to weakly respect the C1-Domination and also fails to weakly respect the C2-Domination by a pair, hence concluding the proof. □

It is worth to note that Translation Invariance is a necessary property of the C1-Correspondences (the beating relation W_p does not change in the translation process). It is also such an extremely weak property of the C2-Correspondences (the difference $p(x, y) - p(y, x)$ does not change in that process), that I do not know of any C2-Correspondence failing to satisfy it. Thus, Proposition 4 shows that all sensible Condorcet C1-Correspondences (those that weakly respect the C1-Domination), and also the wide family of Translation Invariance C2-Correspondences that weakly respect the C2-Domination by a pair, suffer from the Positive and Negative SNSP.

4 Incidence of the Paradoxes in Known Condorcet Correspondences

4.1 C1-Correspondences

As shown by proposition 4, all reasonable Condorcet C1-Correspondences (that is, those weakly respecting the C1-Domination) suffer from the Positive and Negative Strong No Show Paradoxes. In order to see that no Condorcet C1-Correspondence proposed in the literature (as far as I know) is free from these paradoxes, we only need to analyze in detail the **Top Cycle** (f_{TC}) and the **Uncovered Set** (f_{US}) correspondences:

$$f_{TC}(X, p) \equiv \{x \in X\text{: there is no } y \in X \text{ such that } yWW_px \text{ and } (\text{not } xWW_py)\}$$

$$f_{US}(X,p) \equiv \{x \in X\text{: there is no } y \in X \text{ such that } (xW_pz \text{ implies } yW_pz)\}$$

If s is C1-dominated by x, then s is beaten by x, so that s can not belong to $f_{TC}(X,p)$ if x does not. Therefore, f_{TC} weakly respects the C1-Domination and, by Proposition 4, suffers from Positive and Negative SNSP.

On the other hand, if s is C1-dominated by x, then s is covered by x, so that s can not belong to $f_{US}(X,p)$. Therefore, f_{US} respects the C1-Domination and, by Proposition 4, also suffers from these paradoxes. Observe that this is a valid argument for any definition of the covering relation, thus it applies to the correspondences known as **Fishburn**'s function and **Miller's Uncovered set** correspondence, and also to those Uncovered Set correspondences defined in Peris and Subiza (1999) for weak tournaments.

Furthermore, all the others neutral C1-correspondences proposed in the literature are, to our knowledge, refinements of the Uncovered set correspondence, hence they all respect the C1-Domination. This is the case of the following VCs, and also of their counterparts in weak tournaments: **Copeland, Slater, Kendall-Wei, Dutta's Minimal Covering, Banks, Laffond's Bipartisan Tournament Set**, and **Schwartz's Tournament Equilibrium**. See Fishburn (1977), Dutta (1988), Laffond et al. (1993, 1995), Levin and Nalebuff (1995) and Peris and Subiza (1999).

4.2 C2-Correspondences

Proposition 4 shows that all Condorcet C2-Correspondences satisfying a very weak compensation property (that is, those weakly respecting the C2-Domination by a pair) suffer from the Positive SNSP, and that if they also satisfy Translation Invariance, they suffer from the Negative SNSP. All correspondences studied in this section satisfy obviously Translation Invariance.

Let us begin analyzing the **Black** correspondence (f_{BLACK}). See Fishburn (1977).

$$f_{BLACK}(X,p) \equiv \begin{cases} \{c\} & \text{if a Condorcet candidate } c \text{ exists,} \\ \{x \in X\text{: } x \text{ has a maximal Borda Score}\}, & \text{in any other case.} \end{cases}$$

We will show that f_{BLACK} respects the C2-Domination by a pair, because no quasidominated in differences candidate is a winner. If t is quasidominated by x in differences, t can not be the Condorcet candidate and the Borda score of t is lower than that of x. Hence, candidate t is not a winner in $f_{BLACK}(X,p)$. Therefore, f_{BLACK} respects the C2-Domination by a pair, and suffer from Positive and Negative SNSP.

Let us study now the **Kemeny** correspondence (f_{KEM}). See Fishburn (1977), Young and Levenglick (1978), and Young (1995).

$$f_{KEM}(X,p) \equiv \{x \in X\text{: } x \text{ is at the top of an order } L \text{ with maximal Kemeny Score}\}$$

Let (X,p) be a situation with m voters, in which t is C2-Quasidominated in differences by x (being x_s the weak point of x), and $L: x_1 \cdots x_r \cdots x_n$ be an order of X where $x_1 = t$ and $x_r = x$. We will prove that interchanging in L the first candidate t with candidate x, the resulting L_{xt} order has a Kemeny Score higher than that of L.

At the sum $K(L)$ defining the score of L, the only terms not to be present at $K(L_{xt})$ are $p(x_1, x_i), 1 < i \le r$, and $p(x_i, x_r), 1 < i < r$. Let $p(x_i, x_j)$ be called p_{ij}. We have:

$$\begin{aligned} K(L_{xt}) - K(L) &= (\Sigma_{1<i<r} p_{i1} + p_{r1} + \Sigma_{1<i<r} p_{ri}) \\ &\quad - (\Sigma_{1<i<r} p_{1i} + p_{1r} + \Sigma_{1<i<r} p_{ir}) \\ &= (\Sigma_{1<i<r}(m - p_{1i}) + p_{r1} + \Sigma_{1<i<r} p_{ri}) \\ &\quad - (\Sigma_{1<i<r} p_{1i} + p_{1r} + \Sigma_{1<i<r}(m - p_{ri})) \\ &= (p_{r1} - p_{1r}) + \Sigma_{1<i<r} 2(p_{ri} - p_{1i}). \end{aligned}$$

As $x = x_r$ quasi dominates $t = x_1$ in differences, with x_s as the weak point of x_r:

$$\begin{aligned} \text{If } s < r,\ K(L_{xt}) - K(L) &\ge (p_{r1} - p_{1r}) + 2(p_{rs} - p_{1s}) \\ &= (p_{r1} - p_{11} + p_{11} - p_{1r}) + 2(p_{rs} - p_{1s}) \\ &= 2(p_{r1} - p_{11}) - 2(p_{1s} - p_{rs}) > 0 \end{aligned}$$

$$\text{If } s > r,\ K(L_{xt}) - K(L) \ge (p_{r1} - p_{1r}) > 0$$

Hence, in any case, the order L_{xt} has a Kemeny Score higher than that of L, which excludes t from being a winner. Therefore, f_{KEM} respects the C2-Domination by a pair, and suffer from Positive and Negative SNSP.

Now we will study the **Nanson**'s and **Simplified Dogdson**'s correspondences (f_{NAN} and $f_{S.DOG}$). See Fishburn (1977), Richelson (1980) and Young (1995).

$$f_{NAN}(X, p) \equiv lim_{j \to \infty} X_j,\ \textit{where } X_1 = X \quad \textit{and}$$

$$X_{j+1} = X_j, \textit{if all candidates in } X_j \textit{ have the same Borda Score on } (X_j, p/X_j).$$

$$X_{j+1} = X_j \backslash \{x \in X_j : \textit{The Borda Score of } x \textit{ on } (X_j, p/X_j) \textit{ is minimal}\}, \quad \textit{in any other case.}$$

(Observe that the algorithm operates in an iterative fashion, eliminating all candidates with a worst Borda Score in the actual situation, except when all candidates have the same Borda Score)

$$f_{S.DOG}(X, p) \equiv \{x \in X \text{: } x \textit{ needs a minimal number of elemental interchanges in } M_p \textit{ to become a Condorcet candidate}\}$$

If t is quasidominated by x in differences, the Borda score of t is lower than that of x at any step of the elimination process. Hence, candidate t (which will be eliminated before x) is not a winner in $f_{NAN}(X, p)$. On the other hand, the number of elemental interchanges in M_p needed by t to become a Condorcet candidate is obviously higher than that needed by x. This implies that t cannot be a winner in $f_{S.DOG}(X,p)$. Therefore, both f_{NAN} and $f_{S.DOG}$ respect the C2-Domination by a pair, and suffer from Positive and Negative SNSP.

Let us analyze now the **Laffond's Bipartisan Plurality Set** correspondence (f_{BPS}), defined in Laffond et al. (1994). Let (X,p) be any situation in which, as in that of the proof of Proposition 4, $p(x,y) \neq p(y,x)$ for every $x \neq y$.

$$f_{BPS}(X,p) \equiv \{x \in X\text{: } x \textit{ belongs to the support of the unique symmetric Nash Equilibrium of the Bipartisan Plurality Game of } (X,p)\}$$

Laffond et al. (1994) have shown that the Bipartisan Plurality Game of this situation has a unique Nash Equilibrium, and that this equilibrium is symmetric. Let us suppose now that candidate t is C2-Dominated by the pair $\{x,y\}$. If a player of the game plays strategy t with a strictly positive probability ϕ, the best response of the other player can not include t in its support (Indeed, he would obtain a strictly higher payoff by transferring the probability ϕ from t to $0.5x+0.5y$). Therefore, t cannot be in the support of the unique symmetric Nash Equilibrium of the game, and thus t is not a winner in $f_{BPS}(X,p)$. Hence, f_{BPS} respects the C2-Domination by a pair in this type of situations and suffers from Positive and Negative SNSP.

Let us consider now the **Tideman's Ranked Pairs** correspondence (f_{RP}), defined and studied in Tideman (1987).

$$f_{RP}(X,p) \equiv \{x \in X\text{: } x \textit{ is the top candidate of a stack } L\}$$

Proposition 4 is not applicable because f_{RP} does not weakly respect the C2-Domination by a pair nor the C1-Domination (in the situation showed in its proof, the dominated candidate t is the only winner, because the order *tuzyx* is the unique stack). Nevertheless, we will show that f_{RP} suffers from Positive and Negative SNSP. Let $X = \{x,y,z,u\}$ and p the following 15-voters profile: [*uzyx* (4 voters), *xzyu* (3 voters), *yuxz* (3 voters), *xyuz* (2 voters), *zyxu* (1 voter), *zuyx* (1 voter), *xyzu* (1 voter)]. Let v' be the voter from p with preferences L': *zuyx* and v'' be a new voter with preferences L'': *xyuz*. Call p' the profile p with the voter v' removed and p'' the profile $p+v''$. The comparison matrices are:

M_p

	x	y	z	u
x	7.5	6	9	7
y	9	7.5	6	10
z	6	9	7.5	6
u	8	5	9	7.5

$M_{p'}$

	x	y	z	u
x	7	6	9	7
y	8	7	6	10
z	5	8	7	5
u	7	4	9	7

$M_{p''}$

	x	y	z	u
x	8	7	10	8
y	9	8	7	11
z	6	9	8	6
u	8	5	10	8

We will prove that x is elected in (X,p) but not in (X,p') nor in (X,p''). Let us first see that the order $L: xzyu$ is a stack of (X,p). Candidate x attains z through L in (X,p) because $p(x,z) > p(z,x)$, attains y because $p(x,z) \geq p(y,x)$ and $p(z,y) \geq p(y,x)$, and attains u because $p(x,z) \geq p(u,x)$, $p(z,y) \geq p(u,x)$ and $p(y,u) \geq p(u,x)$. Candidate z attains y because $p(z,y) > p(y,z)$, and attains u because $p(z,y) \geq p(u,z)$ and $p(y,u) \geq p(u,z)$. Candidate y attains u because $p(y,u) > p(u,y)$. Therefore, candidate x is a winner.

Let us now see that no order with x at the top can be a stack of (X,p') nor (X,p''). For the order $L: xz_1z_2z_3$ to be a stack for a situation, it is necessary that any candidate in L beats or ties in that situation to his immediate successor. This condition is met in (X,p') only by $L_1: xzyu$ and $L_2: xuzy$. However, $L_1: xzyu$ is not a stack in (X,p') because z fails to attain u through L_1 and $L_2: xuzy$ is not a stack because u fails to attain y through L_2. Exactly the same happens in (X,p''). Thus x is not a winner in (X,p') nor in (X,p'').

The last C2-correspondence to be analyzed, the **Simpson-Cramer Minmax** correspondence (f_{MINMAX}), is (as far as I know) the only one known Condorcet correspondence **free** from SNSP. See Fishburn (1977) and Young (1995).

$$f_{MINMAX}(X,p) \equiv \{x \in X\text{: } \textit{The minimal off-diagonal term of row } x \textit{ in } M_p \textit{ is maximal}\}$$

Note. It is easy to verify that, for $n = 3$ candidates, this rule and the Kemeny rule coincide.

Let us show that f_{MINMAX} satisfies PI and NI. Let (X,p) be any situation and z_1 be a winner candidate for (X,p). Let v' be a voter from p with preferences $L': z_n \ldots z_1$ and v'' be a new voter with preferences $L'': z_1 \ldots z_n$. Call p' the profile p with the voter v' removed and p'' the profile $p + v''$. The only row in $M_{p'}$ whose off-diagonal entries are all the same as that of M_p is the z_1 row (for any row z_i, $p'(z_i,z_j) = p(z_i,z_j) - 1$ if and only if $i < j$). In a similar way, the only row in $M_{p''}$ whose off-diagonal entries are all increased in one unit from that of M_p is the z_1 row (for any row z_i, $p''(z_i,z_j) = p(z_i,z_j) + 1$ if and only if $i < j$). Hence, the z_1 row continues having a maximal minimal off-diagonal entry in $M_{p'}$ and $M_{p''}$. Therefore, z_1 will be a winner for (X,p') and (X,p'').

4.3 C3-Correspondences

We will apply Lemma 1 to prove that the two C3-correspondences proposed in the literature, the **Dogdson** and the **Young** VCs (f_{DOG} and f_{YOUNG}), suffer from Positive SNSP.

$$f_{DOG}(X,p) \equiv \{x \in X\text{: } \textit{The number of elemental interchanges in } (X,p) \textit{ needed by } x \textit{ to become a Condorcet candidate is minimal}\}$$

$$f_{YOUNG}(X,p) \equiv \{x \in X\text{: } \textit{The number of excluded voters in } (X,p) \textit{ needed by } x \textit{ to become a Condorcet candidate is minimal}\}$$

Note. Fishburn (1977) provides slightly different definitions of Dogdson and Young correspondences. The results obtained in this paper are not affected by these modifications.

In the situation (X,p) described in the proof of Proposition 4, since candidate t needs more than 12 elemental interchanges in (X,p) to become a

Condorcet candidate, while y needs only 8 (obtained by switching y and z in 8 voters with preferences *uzytx*), t is not chosen in $f_{DOG}(X, p)$. On the other hand, and for the same situation, since candidate u needs only the removal of 12 voters (the ten voters with preferences *xztyu* and the two with preferences *zyxtu*) to become a Condorcet candidate, while t needs more than 12 voters removed, t is not chosen in $f_{YOUNG}(X, p)$. Therefore, by Lemma 1, both f_{DOG} and f_{YOUNG} fail to satisfy PI and suffer from Positive SNSP.

To prove that f_{DOG} suffers from Negative SNSP, let $X = \{x, y, z, u\}$ and p be the following 17-voters profile: [*uzyx* (5 voters), *zyxu* (5 voters), *xyuz* (4 voters), *uxzy* (2 voters), *yxzu* (1 voter)]. Candidate z needs only 3 elemental interchanges in (X, p) (obtained by switching u and z in 3 voters with preferences *xyuz*) to become a Condorcet candidate, while any other candidate needs more than 3. Hence, z is the only chosen in $f_{DOG}(X, p)$. However, if the four *xyuz* voters are removed from p, u becomes the Condorcet candidate and thus the only chosen. Therefore, f_{DOG} fails to satisfy NI.

Finally, let us show that f_{YOUNG} **is free** from Negative SNSP. Let (X, p) be any situation for which candidate x is a winner (x needs the removal of a minimal number of voters, say k voters, to become a Condorcet candidate), and suppose that the removal of a voter with preference order $L : \ldots x$ causes x to became a loser. This means that in the new situation (X, p') there is a candidate z who, to become a Condorcet candidate, needs the removal of a number h of voters strictly lower than $k - 1$. However, this implies that in (X, p) candidate z would have needed only $h + 1$ voters removed while x would have needed k $(> h + 1)$, contradicting the fact of x being a winner in (X, p).

Therefore, among the VCs studied in this paper, f_{YOUNG} and f_{MINMAX} are the only free from Negative SNSP. It is worth to note that these two rules are in some way symmetrical: f_{YOUNG} (f_{MINMAX}) rule selects candidate x as winner if the number of voters one has to delete from (add to) the initial profile for x to become a Condorcet winner is minimal.

5 Final remarks

Remark 1. *A practical question, which has not been dealt with here, refers to the number of candidates and voters that are necessary to invoke the studied paradoxes. Although a situation with 5 candidates and 39 voters was needed in the proof of Proposition 4, usually a simpler situation (typically of 4 candidates and a number of voters between 15 and 30) is enough to build counterexamples of the Positive and Negative Involvement properties for any given method failing to satisfy them.*

Remark 2 (a stronger and a weaker version of the paradox). *We can define an even stronger no show paradox, called SNSP$^+$, in the following way:*

Definition 11. *A VC f is said to satisfy the* ***Weak Positive Involvement (WPI)*** *property if and only if for any situation (X, p), there is a winner x such that:*

If (X, v) *is a one-voter situation with favorite candidate* x, *then* $x \in f(X, p + v)$.

We say f satisfies the ***Weak Negative Involvement (WNI)*** *property when for any situation* (X, p), *there is a winner* x *such that:*

x remains a winner when we remove from p any voter whose least preferred candidate is x.

A VC f that fails to satisfy any of these properties is said to suffer from SNSP$^+$. The proof of Proposition 4 remains obviously valid for the following alternative statement:

" a) *No Condorcet VC that respects the C1-Domination or respects the C2-Domination by a pair, satisfies the WPI property.*

b) *No Condorcet VC that respects the C1-Domination or respects the C2-Domination by a pair, satisfies the WNI and the Translation Invariance properties"*.

Among the VCs studied in this paper which suffer from some version of SNSP, all suffer from the corresponding stronger version, except the Top Cycle VC and (perhaps) the Tideman's Ranked Pairs VC. In fact, in the situation (X, p) described in the proof of proposition 4, only these two VCs choose candidate t as a winner.

If we allow ties in the voter's preferences, the failure to satisfy the following properties will be a paradox weaker than SNSP, and affecting every Condorcet VC:

Definition 12. *A VC f satisfies the* ***Positive Involvement with ties allowed (PIwta)*** *property if and only if for any given pair of situations* (X, p) *and* (X, v), *where profile v has only one voter,*

If $x \in f(X, p)$ *and x is preferred or tied to any y in v, then* $x \in f(X, p + v)$.

We say f satisfies the ***Negative Involvement with ties allowed (NIwta)*** *property when*

If $x \notin f(X, p)$ *and any y is preferred or tied to x in v, then* $x \notin f(X, p + v)$.

To establish the following proposition, let us make some necessary, but natural modifications in the definition of M_p:

$p(x, y) = p_{STRICT}(x, y) + (1/2)p_{INDIF}(x, y)$, where $p_{STRICT}(x, y)$ is the number of voters who strictly prefer x over y, and $p_{INDIF}(x, y)$ is the number of voters for whom x and y are in a tie. If there are no ties, $p(x, y)$ has the usual meaning.

Proposition 5. *No Condorcet VC f satisfies the PIwta nor the NIwta properties.*

Proof. Let f be a Condorcet VC, $X = \{x, y, z\}$ and p be the following symmetric 15-voters profile $p = [x > y > z$ (1 voter), $y > z > x$ (1 voter), $z > x > y$ (1 voter), $x \sim z > y$ (2 voters), $y > x \sim z$ (2 voters), $x \sim y > z$ (2 voters), $z > x \sim y$ (2 voters), $y \sim z > x$ (2 voters), $x > y \sim z$ (2 voters)$]$. The comparison matrix is:

	x	y	z
x	7.5	8	7
y	7	7.5	8
z	8	7	7.5

Let us suppose, without any loss of generality, that x is a winner. Then, if we add to p two new voters with preferences $x \sim z > y$, or remove from p the two voters with preferences $y > x \sim z$, candidate z becomes a Condorcet candidate and, as a consequence, the only winner. Therefore, f fails to satisfy the PIwta and NIwta properties. □

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